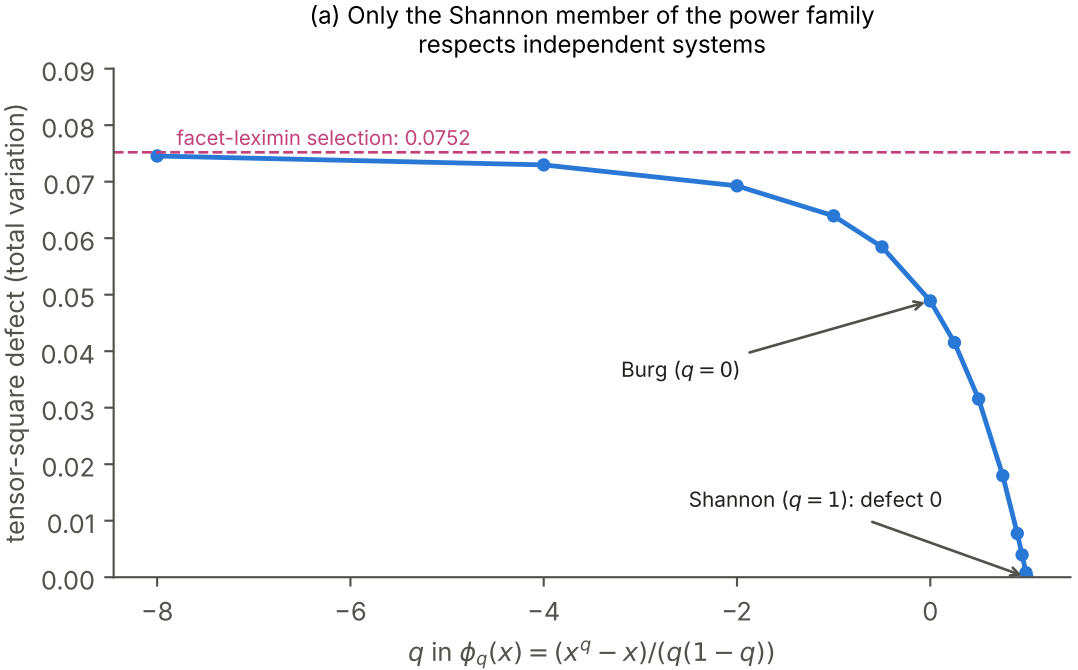
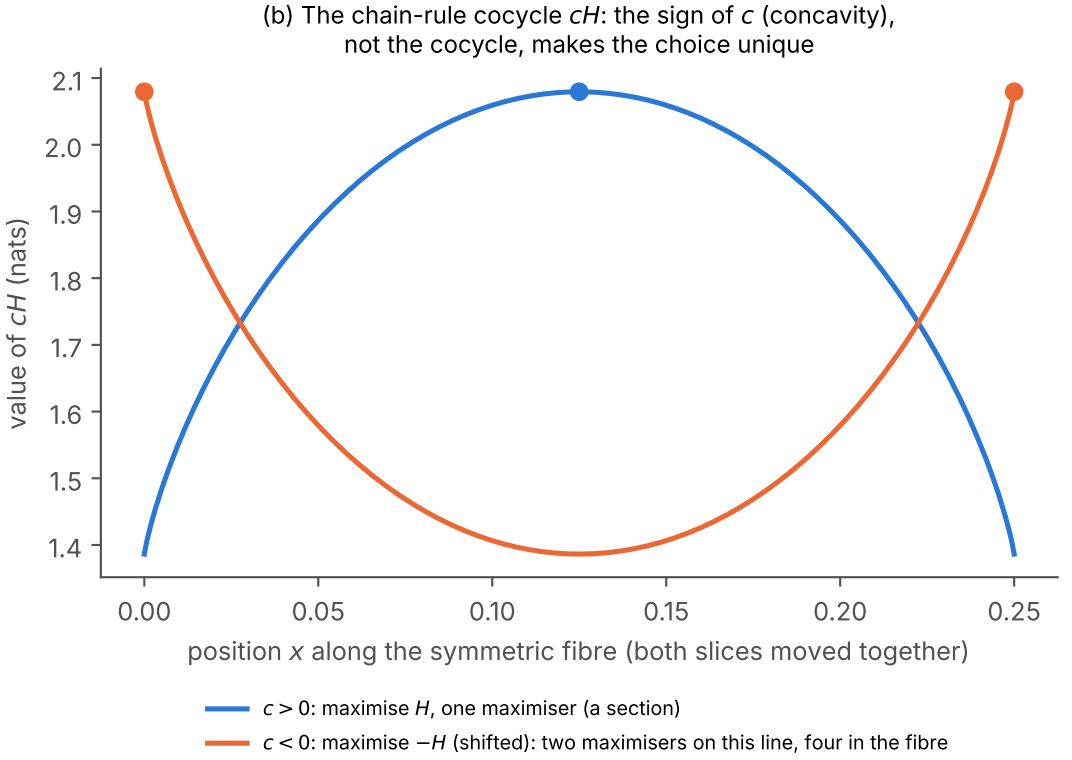

(a) Only the Shannon member of the power family
respects independent systems
facet-leximin selection: 0.0752
Burg ($q = 0$)
Shannon ($q = 1$): defect 0
tensor-square defect (total variation)
0.09
0.08
0.07
0.06
0.05
0.04
0.03
0.02
0.01
0.00
−8
−6
−4
−2
0
q in $\phi_q(x) = (x^q - x)/(q(1-q))$


(b) The chain-rule cocycle cH: the sign of c (concavity),
not the cocycle, makes the choice unique
value of cH (nats)
2.1
2.0
1.9
1.8
1.7
1.6
1.5
1.4
0.00
0.05
0.10
0.15
0.20
0.25
position x along the symmetric fibre (both slices moved together)
$c > 0$: maximise H, one maximiser (a section)
$c < 0$: maximise $-H$ (shifted): two maximisers on this line, four in the fibre